SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA02 **COURSE NAME: C Programming**

COURSE OUTCOME COVERED

CO1: *Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks.* (BL2, BL3)

Assessment Tool Weightage: 10%

QUESTIONS: **Total Marks:100**

Annexure A

MCQ

Code of Conduct:

I M.Nishanth (192511154) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Nishanth

ASSESSMENT TOOL 1

***CO1:** Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)*

Q1. A programmer wants to display the octal equivalent of a decimal number stored as 25 using printf. Which output will the program produce?
int n = 25;
printf("%o", n);
Output?
A) 31
B) 25
C) 32
D) Error

Q2. A developer prints an integer value 26 using the %x format specifier. Considering lowercase hexadecimal representation in C, what will be the output?
int n = 26;
printf("%x", n);
Output?
A) 1A
B) 1a
C) 26
D) Error

Q3. Nerd initializes a variable using 017 and prints it using %d. Since the value is stored in a different number system, what will be the resulting output?
int n = 017;
printf("%d", n);
Output?
A) 17
B) 15
C) 16
D) Error

Q4. A variable is assigned a hexadecimal constant 0xA5 and printed using %d. After decimal conversion, what value will be displayed?
int n = 0xA5;
printf("%d", n);
Output?
A) 155
B) 165
C) 150
D) Error

Q5. Geek uses 010 as the value of an integer and prints it using %x. After converting the value appropriately, what will be the final output?
int n = 010;
printf("%x", n);
Output?
A) 8
B) A
C) 10
D) Error

Q6. A technician adds a hexadecimal value 0x10 and an octal value 010, then prints the result using %d. What will be the output?
printf("%d", 0x10 + 010);
Output?
A) 24
B) 18
C) 20
D) 16

Q7. A variable with value 20 is printed using both %o and %x format specifiers in the same statement. What will be the output?
int n = 20;
printf("%o %x", n, n);
Output?
A) 24 14
B) 20 20
C) 24 20
D) Error

Q8. A macro #define A 10+5 is used in an expression A*2 without parentheses. What will be the output?
#define A 10+5
printf("%d", A*2);
Output?
A) 30
B) 25
C) 40
D) Error

Q9. A macro #define X 10 is defined, and later a variable with the same name int X = 5; is declared. What will happen when printing X?
#define X 10
int X = 5;
printf("%d", X);
Output?
A) 10
B) 5
C) Error
D) Garbage

Q10. A constant variable is declared as "const int x = 10" and then reassigned using "x = x + 5". What will be the result?
const int x = 10;
x = x + 5;
Result?
A) 15
B) 10
C) Error
D) Garbage

Q11. A student divides an integer (5) by a float (2) and prints the result using %f. What will be the output?
int a = 5;
float b = 2;
printf("%f", a/b);
Output?
A) 2.000000
B) 2.500000
C) 2
D) Error

Q12. A float variable is initialized as float x = 5/2 and then printed using %f. What will be the output?
float x = 5/2;
printf("%f", x);
Output?
A) 2.500000
B) 2.000000
C) 5
D) Error

Q13. A character variable is assigned the value 100 and printed using %c. What will be displayed?
char c = 100;
printf("%c", c);
Output?
A) 100
B) d
C) Error
D) Garbage

Q14. A macro, A, is defined and then undefined using #undef. What will happen when attempting to print A?
#define A 5
#undef A
printf("%d", A);
Output?
A) 5
B) 0

C) Error
D) Garbage

Q15. A variable is declared but not initialized, and then printed. What will be the output?
int a;
printf("%d", a);
Output?
A) 0
B) Error
C) Garbage value
D) 1

Q16. A character, 'A', is stored in a variable, and 1 is added to it before printing the result as an integer. What will be the output?
char a = 'A';
int b = a + 1;
printf("%d", b);
Output?
A) 66
B) 65
C) A
D) Error

Q17. A macro #define B 2+3 is used in the expression B + 4. What will be the output?
#define B 2+3
printf("%d", B+4);
Output?
A) 9
B) 10
C) 11
D) Error

Q18. A macro #define X 3+2 is used in the expression X*X without parentheses. What will be the output?
#define X 3+2
printf("%d", X*X);
Output?
A) 25
B) 11
C) 19
D) Error

Q19. An integer is divided by a float and printed using %f. What will be the output of 10/3?
int a = 10;
float b = 3;
printf("%f", a/b);
What will be printed?
A) 3.333333
B) 3.000000

C) 3
D) Error

Q20. A character `A` is printed using %d. What will be the output?
char ch = 'A';
printf("%d", ch);
Output?
A) A
B) 65
C) Error
D) Garbage

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions